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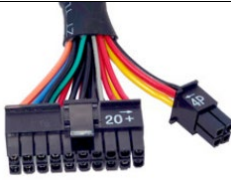
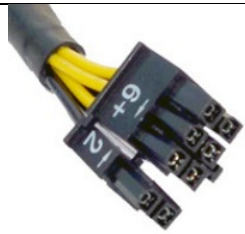
IT Essentials v7.0: Introduction to the Personal Computer

Chapter 1: Online Study Questions



Directions: Use the online curriculum in chapter one to answer the following questions.

Curriculum	Question	Answer
1.1 [1 mark each]	Personal Computer	
1.1.1.1	Watch the Video	<i>"What is in a Computer?" (Try using the CC button at the bottom right to turn on the closed captioning)</i>
	1. What are three main types of components in a PC?	Form Factor
		Power Supply
		Motherboard
	2. Why might there be two networking jacks?	To avoid bad network transmission
	3. What does the CPU do?	Is responsible for interpreting and executing commands.
1.1.2.1	4. Which types of components contain high voltage?	Hard Disk Drives and Fans
	5. Why is electrical equipment grounded?	To prevent electrical shock and direct the flow of electricity to the ground.
1.1.2.2	6. What is the minimum voltage humans can feel?	3000 volts
	7. How much can damage a computer circuit?	8000 volts
	8. What steps should be followed to prevent ESD?	Keep all components in antistatic bags until you are ready to install them.
		Use grounded mats on workbenches
		Use grounded floor mats in work areas
		Use anti-static wrist straps when working inside computers
		[12 marks]
1.2	PC Components	
1.2.1	Case & Power Supplies	
1.2.1.1	9. What are four of the form factors for desktop computer cases?	Horizontal Case
		Full-size Tower
		All-in One Desktop
		Compact Tower

	10. What are some other words used for a computer case?	Tower		
		Cabint		
		Computer Chasis		
		System Unit		
		Box		
1.2.1.2	11. Describe the purpose of a power supply.	Converts alternating-current (AC) wall power into lower voltage direct-current (DC) power		
	12. What type of current comes from the wall socket?	alternating Current (AC)		
	13. What type of current is used by the computer? (it is a lower voltage)	Direct Current(DC). No higher voltage.		
	14. What are some of the form factors for power supplies?	Advanced Technology (AT)		
		AT Extended (ATX)		
		ATX12V		
		EPS12V		
1.2.1.3	15. How are keyed connectors designed to be inserted?	SATA		
	16. Label the following connectors:			
				
				
	17. A keyed connector used to connect to an optical drive.	Molex power connector		
	18. A keyed connector used to connect to a floppy disk drive.	Berg Power connector		
	19. A newer connector used to connect disk drives.	SATA connector		
	20. The main power connector for the motherboard has this many pins:	20-pin or 24-pin slotted connector		

	21. Should cables, connectors or components be forced to fit? Why or why not?	No. Because they fit perfectly with each other.	
	22. Complete the following table for power supplies. (You may need to do some outside research) [5 marks]		
	Voltage	Color	Use
	+12V	Yellow	Disk Drive motors, fans, cooling devices, and system bus slots
	12V	Blue	Some types of serial ports; early PROMs
	+3.3V	Orange	provides the stated voltage
	5V	Red	Motherboard, Baby ATA, earlier CPUs, Mobo Components
	+5V	Purple	Ground
	[Total marks 35]		
1.2.2	Motherboards		
1.1.2.1	23. Which backbone component allows data to travel among the various other system components?	Chipset	
1.2.2.2	24. What are some of the major components attached directly to the motherboard?	CPU	
		RAM Slots	
		BIOS/EUFI	
		Chipset	
		Expansion slots	
1.2.2.3	25. Which part of the chipset controls access to the RAM and video card?	North Bridge	
	26. What does the Southbridge control?	Allows communication for slower speed devices like I/O and storage ports.	
1.2.2.4	27. What term is used to describe the size and shape of the motherboard along with the physical layout of the different components and devices?	Motherboard Form Factor	
	28. Find the following:	Look For	Form Factor
		6.7in x 6.7in	Mini-ITX
		8.5 x 7.5 in	ITX
		Smaller than ATX footprint	Micro-ATX
		Advanced Extended Technology	ATX
	[Total marks 13]		